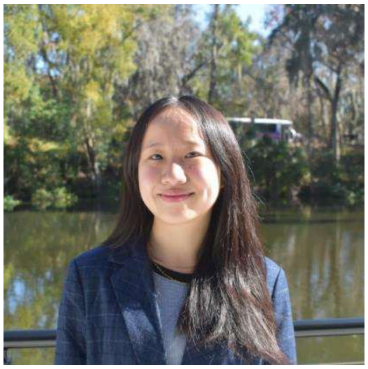

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Methylphenidate and Cue-Evoked Heart Rate in Co-morbid Alcohol Use Disorder and ADHD

Heart rate (HR) increases during exposure to alcohol cues in individuals with alcohol use disorder, reflecting physiological arousal. This study examined whether methylphenidate modulates cue-evoked HR relative to placebo. We hypothesized HR would increase during alcohol cues and that methylphenidate would amplify overall HR and potentially interact with cue type.
Continuous HR time series were derived from filtered cardiac signals and summarized across sliding 5-second windows (1-second step). Linear mixed-effects models accounted for repeated measures and included cue condition, medication, visit, sex, and age as fixed effects, with random intercepts for participants ($n = 33$). Time-course analyses revealed that alcohol cues elicited significantly elevated HR relative to neutral cues ($\beta = +1.00$ bpm, $p = 0.006$). HR declined across the 60-second alcohol cue block, consistent with within-block habituation (Condition × Time interaction: $\beta = -0.024$ bpm per sub-block, $p = 0.031$). Methylphenidate increased HR relative to placebo across conditions ($\beta = +2.39$ bpm, p < 0.001), but no significant Cue × Pill or three-way interactions were observed. These findings highlight the sensitivity of HR dynamics in detecting cue-evoked autonomic responses beyond block averages, supporting their integration with neuroimaging for addiction research.